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ORAL

732. ALLOGENEIC TRANSPLANTATION: DISEASE RESPONSE AND COMPARATIVE TREATMENT STUDIES

Reduced post-transplant alpha-diversity and domination by enterococcus faecium are associated with increased mortality in pediatric recipients of allogeneic hematopoietic cell transplantation.

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Abstract Background: In adult cohorts, gut microbiome features during allogeneic hematopoietic transplant (allo-HCT) period are associated with post-HCT outcomes. Pediatric patients, however, differ substantially in HCT indications, microbiome composition, and immune development, limiting the generalizability of findings from adult studies. Here we hypothesized that gut microbiome diversity and composition are associated with outcomes after HCT in pediatric recipients of allo-HCT.

Methods: We conducted a multi-institutional study enrolling pediatric allo-HCT recipients at Memorial Sloan Kettering (MSK, New York), Ospedale Pediatrico Bambino Gesù (Rome, Italy), and Boston Children's Hospital (BCH, Boston). A total of 349 patients were enrolled, and 1,699 stool samples were collected (30 days pre- through 2 years post-HCT). Samples were spiked for multi-kingdom absolute abundance quantification and shotgun metagenomics was performed.

Results: The median age at transplant was 8.1 years (interquartile range: 4.2–14.1), with 36% female (n=124). Allo-HCT was performed for malignant indications in 66% of patients (n=230). T-cell-depleted allografts were used in 32% (n=110), including α/β T-cell–depleted grafts in 81 patients. Most patients received a bone-marrow graft (n=207, 59%), followed by peripheral blood (n=122, 35%) and umbilical cord blood (n=16, 5%). The majority of patients (n=327, 94%) were undergoing their first allo-HCT; 14 patients underwent a subsequent transplant. Over a median follow-up of 2.6 years (1.7-3.8), 42 patients relapsed (18% of malignant cases), 107 developed acute graft-versus-host-disease (31%), and 59 did not survive (17%), including 22 deaths (6%) from non-relapse mortality. Prophylactic antibiotics varied by site: Rome utilized piperacillin/tazobactam (pip/tazo); MSK initially used ceftriaxone but transitioned to levofloxacin in 2020; and BCH used none.

During the peri-transplant period (day -30 to day +30), microbiome α-diversity, as measured by Simpson reciprocal index, declined significantly post-HCT (median 4.0 pre- vs. 2.1 post-HCT; p<0.001) indicating substantial microbiome injury. Using generalized estimating equations with splines to account for non-linear relationships and adjusting for time, younger age (<3 years), use of T cell-depleted (TCD) graft, and malignant underlying diagnosis were associated with lower a-diversity (p < 0.001). We also observed a significant reduction in median absolute bacterial load, with a nearly 5-fold decrease post-HCT (p< 0.001). Evaluation of institution with principal component analysis (PCA) did not show significant batch effect (p>0.5). In the peri-HCT period, we found that institution and malignant underlying diagnosis were each associated with a significant decrease in bacterial absolute abundance (p<0.001). PCA of the microbiome composition revealed several distinct low-diversity clusters. To further characterize these clusters, we evaluated the incidence of mono-domination, defined as a taxon comprising ≥30% relative abundance. By day +28, any taxon domination was observed in 97% of patients (compared to 4% pre-HCT). Enterococcus was the most frequent dominant taxon (26% of samples), followed by Streptococcus (9%).

To identify clinical predictors of microbiome domination, we used a multivariable generalized linear mixed model (GLMM) with random intercepts. Patients with malignant disease and those treated at the institution with pip/tazo as prophylaxis exhibited higher prevalence of *Enterococcus* domination and higher absolute *Enterococcus* abundance (p <0.001 and p < 0.05, respectively).

Finally, in a multivariable Cox regression analysis, we evaluated the prognostic value of peri-HCT microbiome features among first-time allo-HCT recipients (n=192). In a multivariable analysis, higher peri-engraftment α-diversity (day 7 to day 21) was significantly associated with a reduced risk of all-cause mortality (HR: 0.40, 95% CI 0.17, 0.95; p=0.038). To identify gut microbiome taxa associated with outcomes, we performed FLORAL which identified *Enterococcus faecium* as positively associated with mortality.

Conclusions: This large, multicenter study highlights significant age- and center-specific variability in gut microbiome diversity and composition during pediatric allo-HCT. We found that reduced post-HCT a-diversity and domination by *Enterococcus faecium* were associated with increased mortality.

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